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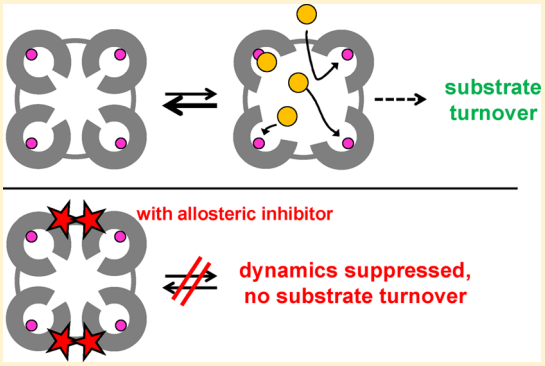
Evidence of Allosteric Enzyme Regulation via Changes in Conformational Dynamics: A Hydrogen/Deuterium Exchange Investigation of Dihydrodipicolinate Synthase

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ABSTRACT: Dihydrodipicolinate synthase is a tetrameric enzyme of the diaminopimelate pathway in bacteria and plants. The protein catalyzes the condensation of pyruvate (Pyr) and aspartate semialdehyde en route to the end product lysine (Lys). Dihydrodipicolinate synthase from *Campylobacter jejuni* (*Cj*DHDPS) is allosterically inhibited by Lys. *Cj*DHDPS is a promising antibiotic target, as highlighted by the recent development of a potent bis-lysine (bisLys) inhibitor. The mechanism whereby Lys and bisLys allosterically inhibit *Cj*DHDPS remains poorly understood. In contrast to the case for other allosteric enzymes, crystallographically detectable conformational changes in *Cj*DHDPS upon inhibitor binding are very minor. Also, it is difficult to envision how Pyr can access the active site; the available X-ray data seemingly imply that each turnover step requires diffusion-based mass transfer through a narrow access channel. This study employs hydrogen/deuterium exchange mass spectrometry for probing the structure and dynamics of *Cj*DHDPS in a native solution environment. The deuteration kinetics reveal that the most dynamic protein regions are in the direct vicinity of the substrate access channel. This finding is consistent with the view that transient opening/closing fluctuations facilitate access of the substrate to the active site. Under saturating conditions, both Lys and bisLys cause dramatically reduced dynamics in the inhibitor binding region. In addition, rigidification extends to regions close to the substrate access channel. This finding strongly suggests that allosteric inhibitors interfere with conformational fluctuations that are required for *Cj*DHDPS substrate turnover. In particular, our data imply that Lys and bisLys suppress opening/closing events of the access channel, thereby impeding diffusion of the substrate into the active site. Overall, this work illustrates why allosteric control does not have to be associated with crystallographically detectable large-scale transitions. Our experiments provide evidence that in *Cj*DHDPS allostery is mediated by changes in the extent of thermally activated conformational fluctuations.



Virtually all biological reactions are catalyzed by enzymes. Rate enhancements of up to 19 orders of magnitude are mediated by active sites that provide specific interactions with substrates, intermediates, transition states, and products.[1,2] In addition to these unique structural features, conformational dynamics appear to be essential for catalytic activity.[3−8] However, the exact manner in which substrate turnover is linked to thermally activated structural fluctuations remains incompletely understood.[3,9]

Cellular homeostasis requires enzyme activity to be carefully controlled via different regulatory mechanisms.[10,11] The noncovalent binding of effector molecules plays a key role in this context. Orthosteric effectors directly target the active site, whereas allosteric effectors bind in locations remote from the active site.[12]

The principles of allosteric control continue to be a matter of debate.[12−15] Most allosteric proteins are multisubunit complexes.[16−20] Early paradigms were developed on the basis of hemoglobin oxygenation,[21−23] culminating in the view that allostery generally involves significant global conformational changes.[14,16] Recent work has placed less emphasis on static structural alterations, focusing instead on the role of dynamic ensembles.[13,14] These newer models envision that enzymes can populate various conformations with different catalytic activities. Within this framework, it is thought that allosteric effectors alter protein energy landscapes, causing population shifts that favor (or disfavor) structures that are catalytically active.[13,14,24] Regardless of the exact mechanism, allosteric effectors somehow modulate the properties of distant active sites.[12−15] Allosteric signal propagation may involve changes in hydrogen bonding, hydrophobic contacts, charge−charge interactions, and entropic factors. Disulfide formation and disruption and local folding and unfolding can play a role, as well.[14,25]

Here we examine a particularly interesting case of allosteric control. Dihydrodipicolinate synthase (DHDPS, EC 4.2.1.52) is required for the biosynthesis of lysine (Lys) via the *m*-diaminopimelate pathway in bacteria and plants. DHDPS

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catalyzes the condensation of pyruvate (Pyr) and aspartate semialdehyde (ASA) to produce dihydrodipicolinate.[26,27] Lys and *m*-diaminopimelate are building blocks of the bacterial peptidoglycan cell wall, making DHDPS a prominent antibiotic target.[26,28,29] Lys acts as an allosteric feedback inhibitor of DHDPS.[27,30,31]

Figure 1a shows the structure of DHDPS from *Campylobacter jejuni* (*Cj*DHDPS),[32] a foodborne Gram-negative

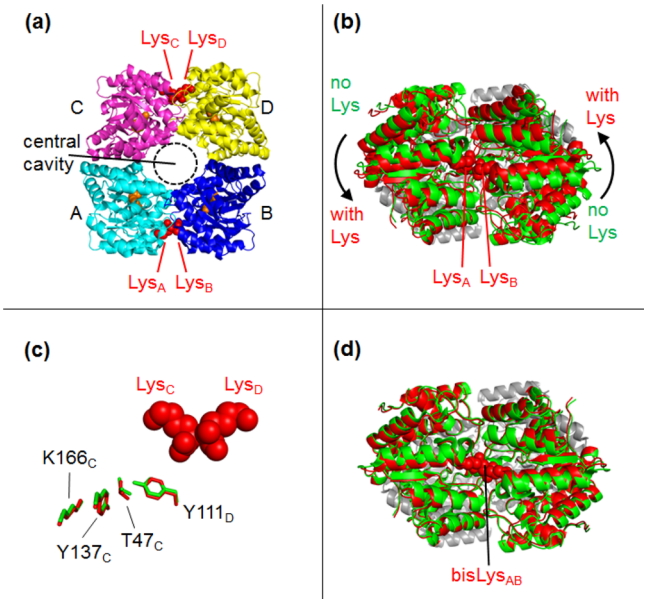


Figure 1. (a) X-ray structure 4M19 of *Cj*DHDPS allosterically inhibited by binding of four Lys residues at the tight interfaces. (b) Overlay of inhibitor-free *Cj*DHDPS (green, 4R53) and Lys-inhibited *Cj*DHDPS (red, 4M19). Lys binding changes the orientation of dimer AB (colored, facing the observer) by ∼4° relative to dimer CD (gray) via counterclockwise rotation. Dimers CD of inhibitor-free and Lys-bound *Cj*DHDPS were aligned with each other in this panel. (c) Orientation of catalytically active residues without inhibitor (green) and in the presence of Lys (red). (d) Overlay of inhibitor-free *Cj*DHDPS (green) and bisLys-inhibited *Cj*DHDPS (red, 5F1V). BisLys binding causes only very subtle alterations in the orientation of dimer AB (colored, facing the observer) relative to dimer CD (gray). Dimers CD of inhibitor-free and bisLys-bound *Cj*DHDPS were aligned with each other.

pathogen.[33] The homotetrameric complex exhibits a "dimer of dimers" architecture, where subunits A and B (as well as C and D) form tight interactions. Contacts at the A−C and B−D interfaces are less extensive. The four subunits surround a water-filled central cavity. Allosteric binding takes place at the tight interfaces. Inhibited *Cj*DHDPS accommodates two pairs of Lys molecules, causing the two tight dimers to rotate against one another by ∼4° (Figure 1b). Subtle tertiary structural changes take place, as well, but the orientations of key active site residues (K166, as well as the catalytic triad[34] of Y137, T47, and Y111′) remain virtually unchanged upon inhibitor binding (Figure 1c).[32]

The mechanism whereby Lys binding allosterically inhibits *Cj*DHDPS (and other DHDPS variants) remains unclear.[29,30,32] Subunit rotations qualitatively similar to those displayed in Figure 1b are quite common in allosteric enzymes.[16,17,21−23] However, changes in the quaternary and tertiary structure of other allosteric systems tend to be much larger than for *Cj*DHDPS.[17,29,32] Crystallography revealed a slight expansion of the active site cavity upon Lys binding,[32] while molecular dynamics (MD) simulations indicate diminished side chain dynamics for the active site tyrosines.[30] Both of these effects may contribute to inhibition, but it is not obvious how either of them could be a key factor in allosteric control.

Recent work has identified a potent novel *Cj*DHDPS inhibitor.[29] This synthetic compound, termed bisLys, consists of two Lys molecules that have their C_α atoms linked via an ethylene bridge. BisLys targets the allosteric site in a fashion that mimics the binding of two Lys molecules, but it does so with a 300-fold higher affinity. This tighter binding is consistent with entropic effects previously demonstrated for other systems.[35,36] Conformational alterations caused by bisLys are even smaller than those seen upon Lys binding (Figure 1b,d). The rmsd values relative to the inhibitor-free enzyme are 0.456 and 1.395 Å, respectively. These data further reinforce the view that static structural changes are not a main contributor to allosteric inhibition of *Cj*DHDPS.[29]

The preceding discussion points to the possibility that allosteric control of *Cj*DHDPS is based on changes in structural dynamics, rather than switching from an active to an inactive conformation.[32] Catalysis proceeds via a ping-pong mechanism, with Schiff base formation between Pyr and K166 as the first step. This is followed by ASA binding and aldol condensation.[27] K166 is located deep within the protein (Figure 2a). X-ray data

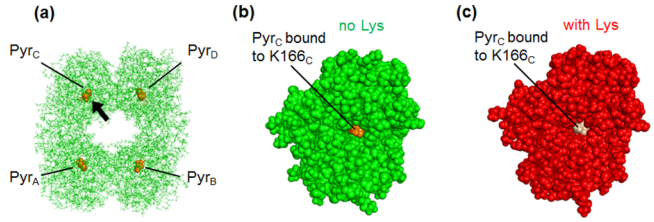


Figure 2. (a) Substrate (Pyr) position in the active site (PDB entry 4LY8). The black arrow indicates the orientation of the substrate access channel from the central cavity. (b) View from the central cavity toward the active site of subunit C along the substrate access channel in the absence of Lys (PDB entry 4R53 with hydrogens added using Pymol). (c) Same as panel b, but after Lys binding (PDB entry 4M19).

suggest that Pyr gains access to K166 from the central cavity through an ∼10 Å long channel (black arrow in Figure 2a). This conduit provides the most unobstructed view of the partially buried substrate in the crystal structure (Figure 2b,c). However, this access channel is barely wide enough to allow passage of Pyr. A static bottleneck of this type would be incompatible with efficient catalysis, as it would restrict diffusion of the substrate into the active site. Catalysis must therefore involve transient "opening" events that visit conformations where K166 is more accessible than in the crystal structure. Such a dynamic access scenario is consistent with experimental data[37] and MD simulations[38] on other proteins with buried binding sites, and it reflects general views of the functional role of enzyme motions.[3−7] Our considerations highlight a possible solution to the conundrum of *Cj*DHDPS allosteric regulation. Specifically, inhibitor binding may suppress catalytically relevant fluctuations,[32] including those required for binding of Pyr to K166. In summary, we hypothesize that (i) *Cj*DHDPS fluctuations are essential for substrate turnover and (ii) allosteric inhibitors act by interfering with these fluctuations.

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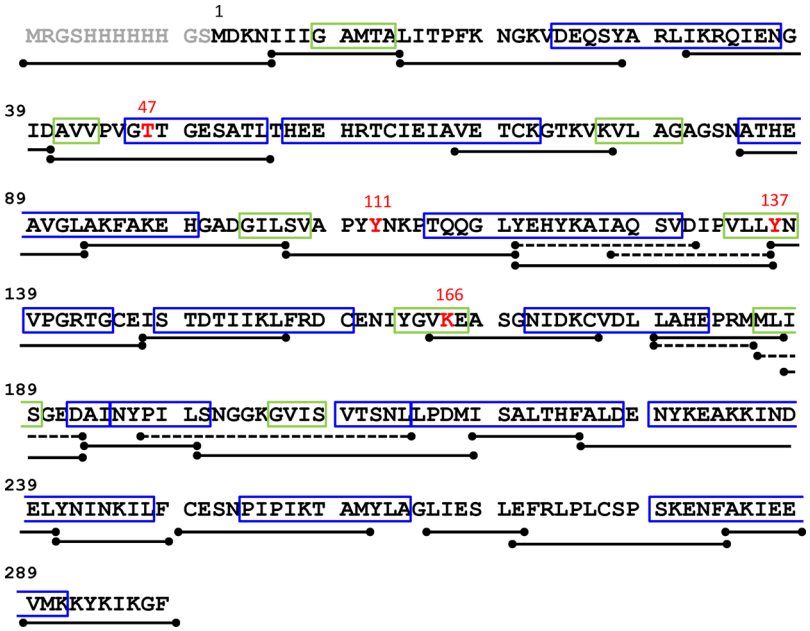


Figure 3. Sequence of the wild-type *Cj*DHDPS construct used in this work. The leading His tag is colored gray, and the residue numbering starts with the N-terminal methionine as in the corresponding crystallographic data. Solid lines indicate the peptic peptides used for HDX data analysis. Dashed lines represent redundant peptides. Key active site residues are colored red, α-helices blue, and β-strands green.

This work employs hydrogen/deuterium exchange (HDX) mass spectrometry (MS) for monitoring changes in *Cj*DHDPS dynamics upon inhibitor binding. Our aim is to improve the understanding of *Cj*DHDPS allosteric inhibition by scrutinizing the viability of hypotheses (i) and (ii) mentioned above. The HDX method is a well-established tool for probing protein fluctuations. Backbone deuteration is mediated by the transient disruption of N−H···O═C hydrogen bonds, such that dynamic regions exhibit HDX rates higher than those of rigid segments.[24,39−42] Earlier studies used HDX MS for probing allosteric effects in other proteins,[24,43−48] but it appears that there have been no prior HDX MS investigations of *Cj*DHDPS. We find that inhibitor-free *Cj*DHDPS exhibits a distinctive deuteration behavior, consistent with the view that substrate access is mediated by large-scale structural fluctuations. Inhibitor binding suppresses deuteration in a site-selective fashion, clearly defining allosteric elements that undergo rigidification. Changes observed in the vicinity of the substrate access channel support a direct relationship between reduced conformational dynamics and allosteric inhibition.

MATERIALS AND METHODS

Materials. 4-(2-Hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), L-lysine, and glycerol were purchased from Sigma (St. Louis, MO). Sodium chloride was from Caledon (Georgetown, ON), and D_2O was from Cambridge Isotope Laboratories (Andover, MA). Wild-type *Cj*DHDPS (tetramer mass of 136277 Da) was overexpressed in *Escherichia coli* and isolated following established procedures.[32] BisLys was synthesized as described previously.[29]

Hydrogen/Deuterium Exchange Mass Spectrometry. Experiments started with stock solutions of 80 μM *Cj*DHDPS in 50 mM HEPES and 100 mM NaCl with 10% glycerol (pH 7.2). BisLys-containing samples were prepared by adding 1 μL of 12 mM bisLys in water to 10 μL of a protein solution, resulting in a 15-fold molar excess of inhibitor. Lys-bound samples were prepared by addition of 1 μL of 1 M Lys in water to 10 μL of a protein solution, resulting in a 1200-fold molar inhibitor excess. The solutions were then diluted to a total volume of 40 μL using aqueous buffer. These mixtures were pre-equilibrated for 30 min at 4 °C. Deuteration was conducted at room temperature (22 ± 1 °C). HDX was initiated by addition of 9 volumes of D_2O-based labeling buffer that had the same salt and buffer composition as the stock solution, for a final measured pH of 7.2, a protein concentration of 2 μM, and inhibitor concentrations of 30 μM (bisLys) or 2.4 mM (Lys). For K_i values around 0.2 and 60 μM, respectively, the fraction of bound protein under HDX conditions was estimated to be ∼99%;[29] 30 μL aliquots were removed at various time points ranging from 1 to 120 min after the initiation of labeling. These aliquots were quenched at pH 2.4 by addition of HCl on ice, followed by flash-freezing in liquid nitrogen and storage at −80 °C. For spatially resolved HDX MS experiments, the aliquots were rapidly thawed to ∼0 °C and manually injected into a nanoACQUITY UPLC with HDX technology (Waters, Milford, MA).[49] Online digestion was performed on a POROS pepsin column (2.1 mm × 30 mm) from Life Technologies/Applied Biosystems (Carlsbad, CA) at 15 °C. Desalting and peptide separation were performed within 15 min on a 1.7 μm BEH C18 reversed phase column (1 mm × 100 mm) using a water/acetonitrile gradient in the presence of 0.1% formic acid at a rate of 35 μL min^{-1}. The temperature for peptide trapping and C18 separation was set to 0 °C. Blank runs between protein injections ensured the absence of sample carryover. This procedure yielded a total of 34 peptides with signal/noise (S/N) ratios that were adequate for reliable HDX MS measurements. The subsequent considerations will focus on a subset of 27 peptides that cover the protein in a contiguous fashion with a sequence coverage of 96% (Figure 3).

Mass analysis of peptides was performed on a Waters Synapt G2 HDMS instrument with source and desolvation temperatures of 80 and 250 °C, respectively, a cone voltage of 20 V, and an electrospray voltage of 2.8 kV. The identity of each peptide was confirmed by tandem MS based on the known

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*Cj*DHDPS sequence. Zero-time point controls (m_0) for the correction of in-exchange were performed by exposing DHDPS to quenching buffer, followed by D_2O exposure, resulting in the same final solution composition that was observed for all other samples. Controls for fully exchanged DHDPS (m_{100}, for the correction of back exchange) were prepared by incubating 20 μM DHDPS in a labeling solution at pH 2.0 for 1 day. BioLynx version 4.1 (Waters) and HX-Express[50] were employed to analyze the centroid mass of all peptides as a function of labeling time. Deuteration levels (% deuteration) were determined as

$$\%\ \text{deuteration} = \frac{m - m_0}{m_{100} - m_0} \times 100\% \quad (1)$$

and average deuteration differences were calculated as

$$\%\ \Delta\text{deuteration} = \frac{1}{N}\sum (\%\ \text{deuteration}_{\text{ligand-bound}} - \%\text{deuteration}_{\text{without ligand}}) \quad (2)$$

The sum in eq 2 extends over the $N = 5$ time points that were measured for each peptide. All measurements were performed in triplicate. Error bars represent standard deviations. Solid lines in HDX MS plots represent mono- or biexponential fits that were included for the purpose of illustration.

Intact protein HDX MS was conducted without using the integrated HDX module. The injection syringe, column, injector, and solvent delivery lines were kept at 0 °C in an ice bath. Experiments were started by equilibrating the protein with bisLys or Lys in aqueous buffer. Deuterated labeling buffer was then added to initiate the exchange reaction as described above. Quenching of aliquots was performed by the addition of cold HCl to decrease the pH to 2.4. The quenched solutions were loaded on a C4 column (BEH300 C4, 1.7 μm, 2.1 mm × 50 mm) at a flow rate of 200 μL min^{-1} for desalting and mass analysis. The injection loop volume was 20 μL, and the total protein amount per injection was 40 pmol. For HDX kinetic measurements, labeling was conducted at constant inhibitor concentrations, and aliquots were analyzed at various time points. For PLIMSTEX[51] measurements, a constant labeling time of 60 min was used for all samples, while incubation was conducted at different inhibitor concentrations.

RESULTS AND DISCUSSION

Global HDX Measurements. To examine the interactions of *Cj*DHDPS with its allosteric inhibitors Lys and bisLys, we initially conducted HDX MS experiments at the intact protein level. PLIMSTEX[51] measurements probe the deuteration behavior at a constant HDX time point, while the inhibitor concentration is varied. We chose 60 min, which reflects a regime in which the protein has undergone extensive labeling but is still far from being fully deuterated. For inhibitor-free *Cj*DHDPS, 64% of the amide backbone hydrogens become deuterated under these conditions. With an increase in the Lys concentration, the PLIMSTEX profile drops rapidly, until it levels off at a deuteration level of 47% for Lys concentrations around 200 μM (Figure 4a). Similar effects were observed in the presence of bisLys, except that the PLIMSTEX profile plateaued at concentrations roughly 2 orders of magnitude lower than those for Lys (Figure 4b). All subsequent HDX measurements were performed with saturating inhibitor concentrations, i.e., 2.4 mM for Lys and 30 μM for bisLys

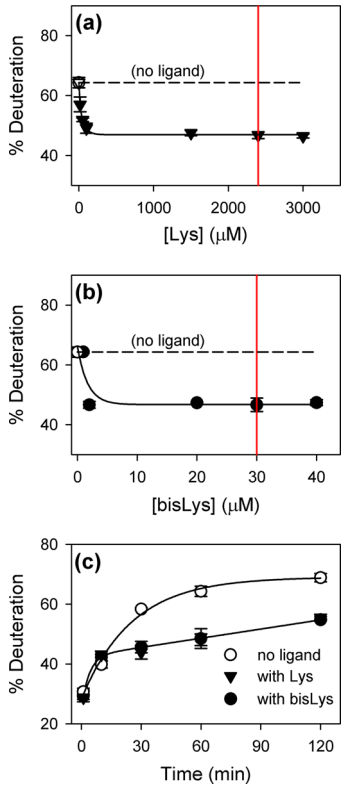


Figure 4. Effects of binding of an allosteric ligand to *Cj*DHDPS monitored by intact protein HDX MS. (a) Deuteration level at different concentrations of Lys, monitored after HDX for 60 min. (b) Same as panel a, but for bisLys binding. Note the different concentration range used for the two panels. Red vertical lines indicate concentrations used for subsequent HDX MS measurements. (c) Deuteration kinetics monitored as a function of time at constant inhibitor (Lys and bisLys) concentrations. Also included are data acquired in the absence of inhibitor.

(vertical lines in Figure 4a,b). Time-dependent data acquired with these inhibitor-saturated solutions resulted in very similar global HDX kinetics for Lys and bisLys. In contrast, greatly enhanced deuteration was observed for inhibitor-free *Cj*DHDPS (Figure 4c).

The data of Figure 4 reveal that binding of either Lys or bisLys significantly reduces the conformational dynamics of *Cj*DHDPS; i.e., the transient disruption of backbone hydrogen bonds is less extensive for the allosterically inhibited protein. Much higher concentrations of Lys are required to achieve this effect than in the case of bisLys, consistent with the lower binding affinity of the former.[29] In general, binding of a ligand to allosteric proteins can elicit various responses, ranging from strongly stabilizing effects to global destabilization.[44] Binding of Lys and bisLys to *Cj*DHDPS causes a net stabilization, evident from the reduced deuteration levels seen in the presence of either inhibitor.

Spatially Resolved HDX MS. A more detailed view of the *Cj*DHDPS dynamics is obtained upon incorporation of proteolytic digestion and peptide mapping into the HDX MS workflow. The resulting spatially resolved deuteration kinetics reveal a wide spectrum of behaviors for the various protein regions (Figure 5). Some segments (such as amino acids 67–76) maintain deuteration levels close to zero regardless of experimental conditions, implying that structural fluctuations with transient H-bond disruptions are rare in these regions.

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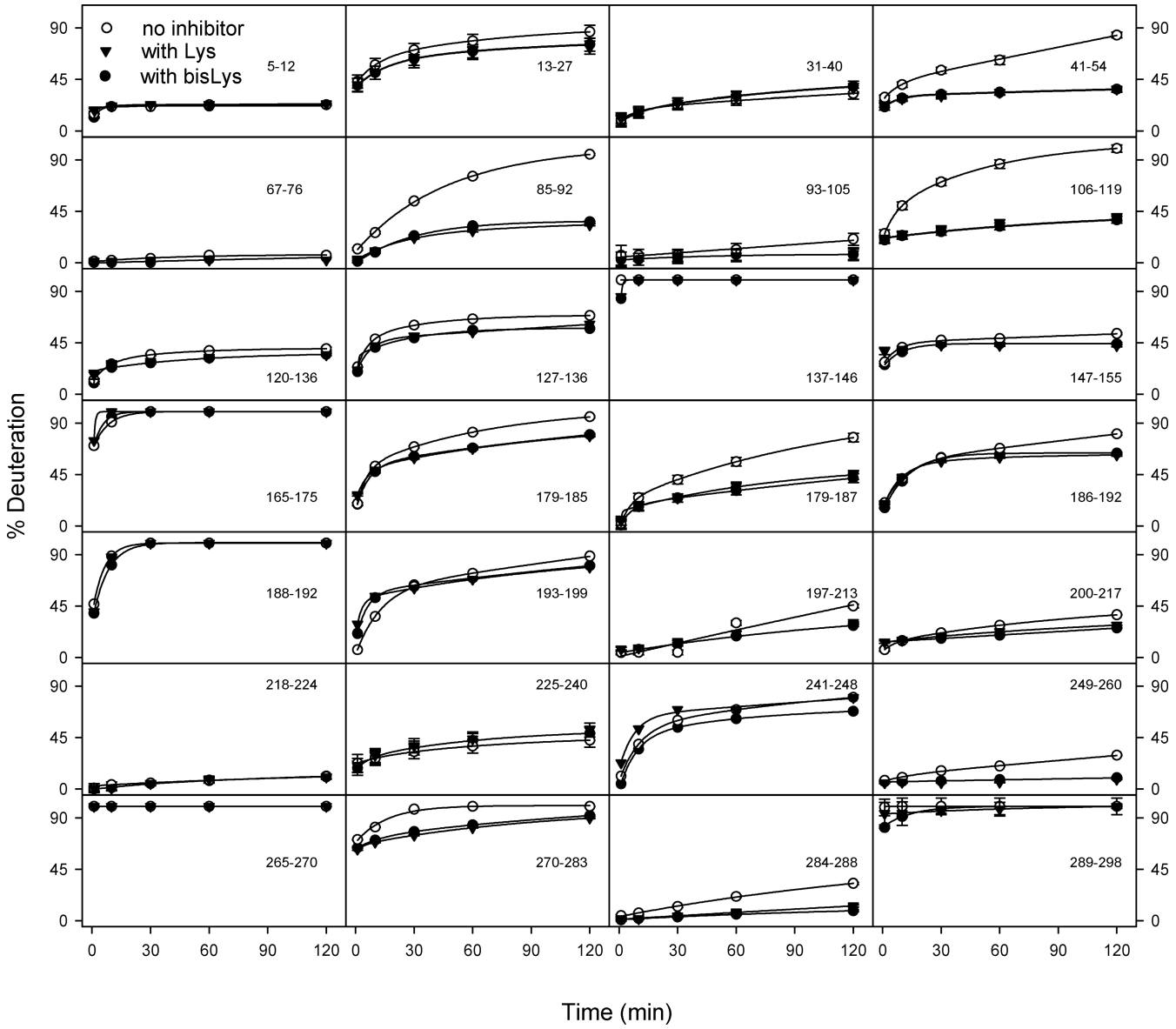


Figure 5. Summary of *Cj*DHDPS deuteration kinetics measured for the various peptic peptides. Each panel contains data for three experimental conditions: without inhibitor (○), with Lys (▼), and with bisLys (●). The sequence range covered by each of the peptides is indicated, using the residue numbering defined in Figure 3.

Other segments show deuteration levels close to 100% already at the earliest time point, reflecting a high degree of protein dynamics and/or the absence of stable backbone H-bonds (exemplified by the C-terminal peptide of residues 289−298). Addition of an inhibitor dramatically reduces the level of deuteration for some segments (residues 41−54, 85−92, and 106−119), while others are much less affected by binding of Lys or bisLys (such as residues 31−40). By and large, the deuteration responses elicited by the two inhibitors are quite similar to one another, although some differences are noticeable in the range of residues 241−248.

Crystal Structures and Deuteration Patterns. X-ray crystallography provides an indispensable foundation for the interpretation of HDX data. It is important to remember, however, that X-ray structures depict static protein conformations in the solid state under cryogenic conditions.[29] They "show the stage but not the action".[52] In contrast, HDX kinetics represent the behavior in native solution at ambient temperature where proteins undergo a wide range of thermally activated motions.[8,39] Although the average conformations in these two environments can be expected to be quite similar, there is no guarantee that each feature seen in X-ray structures is fully retained in solution. For example, some helices that are well developed in the crystal may be only marginally stable under solution conditions.[53] With these caveats in mind, we will now map the deuteration levels for 60 min (from Figure 5) to the corresponding *Cj*DHDPS X-ray structures.

Conformational Dynamics of Inhibitor-Free *Cj*DHDPS. Figure 6a illustrates the HDX pattern of inhibitor-free *Cj*DHDPS. Red and orange represent high deuteration levels, as defined at the bottom of the figure. The prevalence of these two colors in Figure 6a attests to the highly dynamic nature of the inhibitor-free enzyme. Interestingly, the most dynamic regions border the central cavity of the tetramer (red in Figure 6a). Many of the helices in the protein periphery are much more rigid, evident from their blue and green coloring. This

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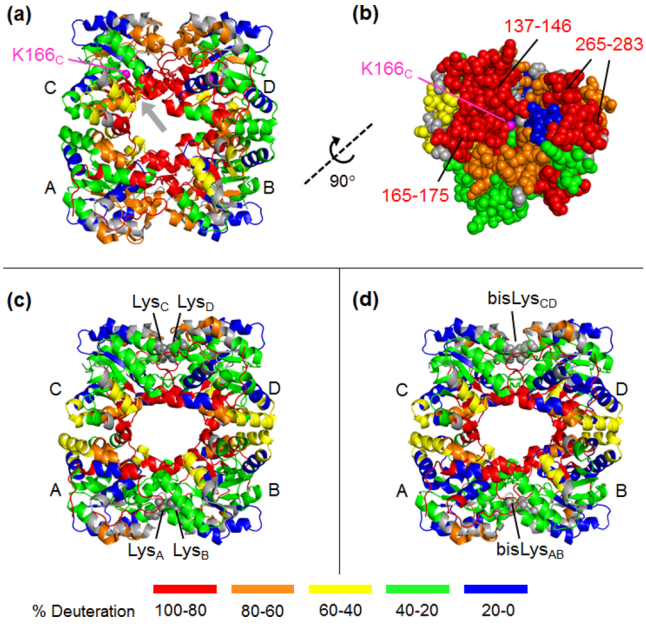


Figure 6. Deuteration percentages for 60 min (from Figure 5) mapped to the crystal structures of *Cj*DHDPS. (a) No ligand added (PDB entry 4R53). (b) View toward the active site of subunit C from the central cavity, as indicated by the gray arrow in panel A. Heavy atoms are shown in space-filling representation. N_ε of the catalytic site K166 is colored magenta. (c) HDX pattern after Lys binding (PDB entry 4M19). (d) HDX pattern after bisLys binding (PDB entry 5F1V).

pattern is quite different from the HDX properties of other tetrameric proteins such as hemoglobin, where peripheral helices are among the most dynamic.[44]

The fact that the *Cj*DHDPS central cavity is surrounded by particularly dynamic segments likely represents a functionally relevant feature of the protein architecture. As outlined in the introductory section, static crystallographic data[32] make it difficult to envision how substrate molecules can gain access to the occluded active site K166. Our HDX MS data strongly suggest that this impediment is overcome via extensive conformational fluctuations that provide transient access to the active site. This interpretation is reinforced in Figure 6b, which provides a view from the central cavity along the narrow substrate access channel toward the active site K166. It is seen that this region comprises some of the most dynamic segments [residues 137−146, 165−175, and 265−283 (red in Figure 6b)]. Even the catalytic residue K166 seems to be highly dynamic, which is evident from the very rapid deuteration of the segment of residues 165−175 (Figure 5). The same goes for R142, which is involved in ASA binding.[54] These HDX MS data are consistent with the general paradigm[3−8] that structural dynamics are an essential component of enzyme function. Specifically, extensive opening/closing fluctuations of *Cj*DHDPS appear to play a key role in providing access of the substrate to the active site. This conclusion supports the viability of hypothesis (i) stated in the introductory section.

Inhibitor-Induced Changes in Conformational Dynamics. Lys- and bisLys-inhibited *Cj*DHDPS show reduced deuteration levels in several regions (Figure 6c,d) compared to those of the inhibitor-free protein (Figure 6a). In particular, segments around the allosteric sites at the tight subunit interfaces undergo marked stabilization upon inhibitor binding. This effect is attributed to crystallographically detected hydrogen bonds between the inhibitors and the protein.[32] Deuteration levels at the loose A−C and B−D interfaces are somewhat enhanced, indicating a slight destabilization of these regions in the presence of inhibitor (Figure 6c,d).

HDX difference maps provide a clearer perspective of the inhibitor-induced changes (Figure 7a,c). The regions that undergo the most pronounced rigidification [residues 41−54, 85−92, and 106−119 (see also Figure 5)] are in direct contact with the allosteric binding sites. These segments also extend into the interior, and two of them carry active site residues [T47 and Y111 (Figure 1c)]. It is quite possible that reduced structural dynamics at these two residues contribute to allosteric inhibition of *Cj*DHDPS. Unfortunately, the limited spatial resolution of our data makes it difficult to determine the extent to which rigidification of the allosteric site is transmitted to T47 and Y111. Also, it is not known whether structural fluctuations of these particular residues are a prerequisite for catalysis. Consequently, it seems advisable to scrutinize the HDX difference maps for additional clues about the inhibition mechanism.

Panels b and d of Figure 7 show HDX difference maps seen from the central cavity toward the active site, along the substrate access channel. Large segments bordering the central cavity become stabilized upon Lys or bisLys binding [residues 249−260 and 270−288 (blue in panels b and d of Figure 7)]. Stabilization of these regions clearly highlights the allosteric nature of Lys and bisLys binding, because they are not in direct contact with the inhibitors. Instead, both of these regions are packed against the segment of residues 41−54, which is part of the allosteric binding site. This arrangement implies that residues in the range of 41−54 participate in allosteric signal transmission.

Is it possible that the observed rigidification of segments of residues 249−260 and 270−288 is important for allosteric inhibition of *Cj*DHDPS? We noted previously how access of Pyr to the buried K166 appears to be a limiting factor for substrate turnover, and we outlined how conformational fluctuations around the substrate access channel can help overcome this problem. Rigidification of residues 249−260 and 270−288 will likely interfere with such functionally required motions. In particular, the segment of residues 249−260 comprises residues that line the walls of the substrate access channel. Inhibitor-induced stability enhancements in this region will help ensure that the access channel permanently maintains the narrow diameter that is seen in the crystal structures. The resulting steric bottleneck will impede diffusion of the substrate to the active site, thereby reducing the overall reaction rate. The proposed scenario establishes a simple link between the conformational dynamics and enzymatic activity of *Cj*DHDPS, consistent with hypothesis (ii) outlined in the introductory section.

Considering the differences in the chemical nature of the two inhibitors studied here, the possibility that Lys and bisLys would elicit exactly the same changes in protein dynamics would not be expected. Lys causes a subtle destabilization of the segment of residues 241−248 (Figure 7a,b), while this region is slightly rigidified in the presence of bisLys (Figure 7c,d). It is possible that this difference is partly responsible for the fact that bisLys represents a more potent inhibitor of *Cj*DHDPS,[29] keeping in mind that the region of residues 241−248 may be involved in gating access of the substrate to the active site.

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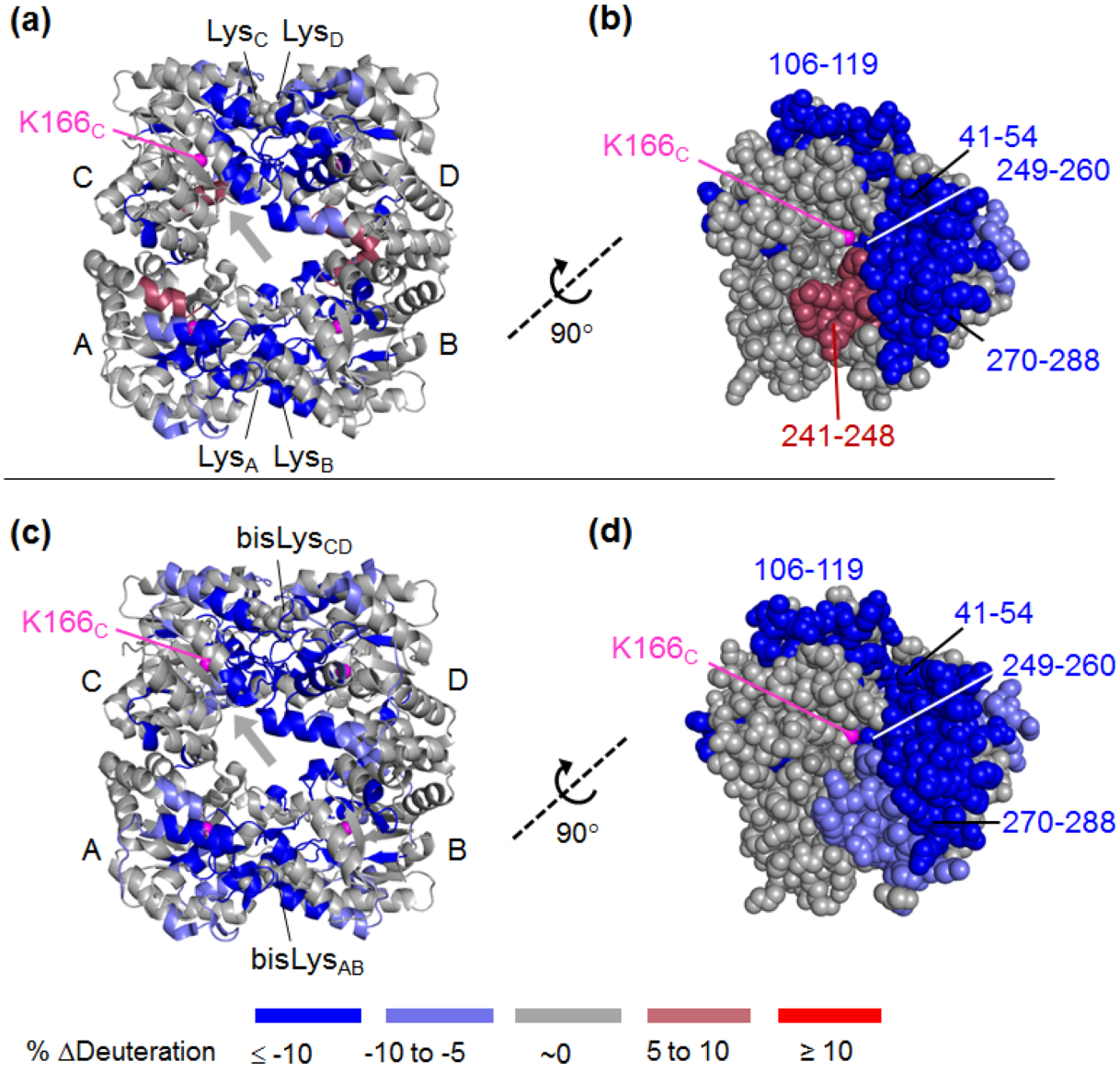


Figure 7. Deuteration difference plots mapped to *Cj*DHDPS crystal structures. Regions with lower deuteration levels after ligand binding are colored blue, as indicated in the color scheme at the bottom. (a) Deuteration changes upon Lys binding. (b) Same as panel a, viewed from the central cavity for subunit C in space-filling representation. (c and d) Corresponding deuteration changes after bisLys binding.

CONCLUSIONS

It is widely accepted that structural fluctuations are required for enzyme catalysis,[3−7] but the causative link between these random motions and protein activity is not always obvious.[8] In the case of *Cj*DHDPS, the mechanistic role of thermally activated opening/closing events appears to be relatively easy to understand. In the absence of such fluctuations, it would be difficult to envision how the substrate (Pyr) can reach the buried active site residue K166. The HDX data of this work are consistent with a dynamic access mode, supporting the viability of hypothesis (i) outlined in the introductory section (Figure 8a). It can also be envisioned that binding of the second substrate (ASA), as well as release of the dihydrodipicolinate product, will depend on protein structural fluctuations. Whether catalytically required opening/closing events in *Cj*DHDPS take place cooperatively or whether they affect individual subunits independent of one another remains unknown.

Our data also support hypothesis (ii) that states that allosteric inhibitor binding suppresses enzyme activity by interfering with catalytically relevant fluctuations, in particular those that are required for access of the substrate to the active site (Figure 8b). In addition, it is possible that the inhibitor-induced rigidification of the enzyme seen in our HDX experiments also affects other dynamic motions involved in substrate turnover.[30,32] In any case, these considerations provide an explanation for the seemingly perplexing finding[29,32] that allosteric inhibition of *Cj*DHDPS takes place without crystallographically detectable large-scale changes. Activity of this protein is controlled at the level of conformational dynamics, rather than modifications of its average structure.

The focus of the preceding discussion is the diffusion of Pyr to K166 via the substrate access channel from the central cavity. The available crystallographic information suggests that this channel offers the least obstructed path from bulk solution to the active site. Studies of other proteins indicate that structural fluctuations can sometimes allow substrate diffusion along

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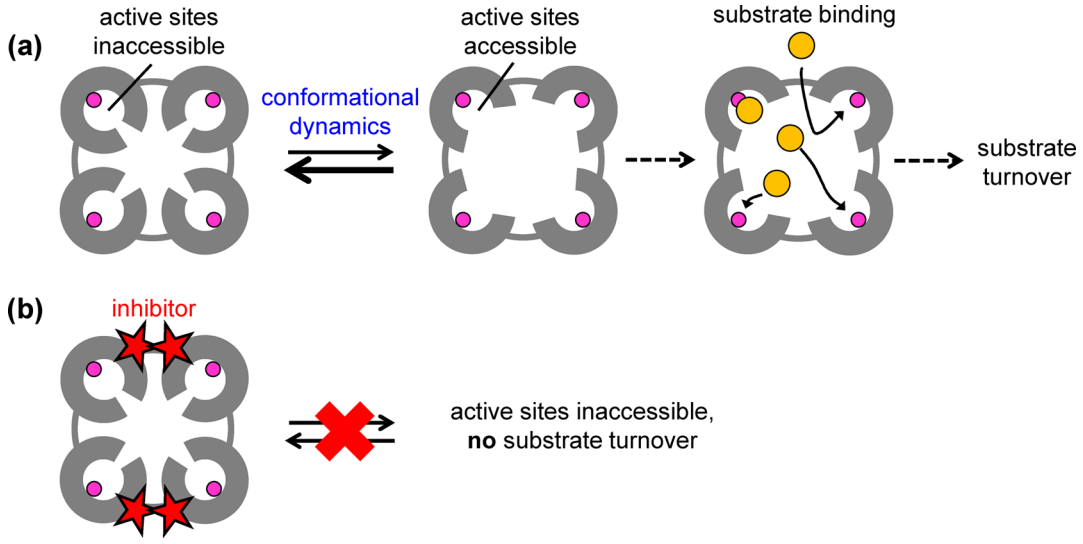


Figure 8. Proposed mechanism of *Cj*DHDPS allosteric inhibition. (a) No inhibitor present. From left to right, the cartoon shows the following. (1) The ground state of the tetramer in solution, where active site access is sterically blocked. (2) Thermally activated structural fluctuations provide access to active sites. (3) The substrate diffuses into the active sites while they are transiently accessible; this is followed by substrate turnover. Opening/closing transitions may take place independently for each subunit, or in a cooperative fashion. (b) Inhibitor binding suppresses structural fluctuations. The substrate cannot gain access to active sites due to "stiffening" of the enzyme, which prevents the access channel from transiently opening. As a result, catalytic turnover is shut down. Pink circles represent key active site residues, such as K166 in *Cj*DHDPS.

multiple pathways, some of which may not be readily apparent from static X-ray data.[38] In other words, the access channel discussed here may not be the only viable route by which Pyr can reach K166. Other routes may exist, perhaps accessing K166 from the "outside" of the tetramer. Very likely, such alternative access routes will also be inhibited by the HDX MS-detected rigidification, such that the overall inhibition mechanism proposed in Figure 8b remains viable.

This study reaffirms the role of HDX MS as a valuable tool for deciphering allosteric aspects of protein regulation.[24,43−48] In future work, it will be interesting to further explore the role of enzyme dynamics during turnover and inhibition by complementing HDX MS experiments with MD simulations on substrate diffusion at the water−protein interface and in the protein interior. Earlier conformational sampling simulations focused largely on small molecules such as diatomic gases[38] and water.[55] It is our hope that similar approaches can be extended to bioorganic substrates and products of typical enzymatic reactions.

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Notes

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ABBREVIATIONS

ASA, aspartate semialdehyde; bisLys, synthetic high-affinity inhibitor (bis-lysine); *Cj*DHDPS, dihydrodipicolinate synthase from *C. jejuni*; HDX, hydrogen/deuterium (H^1/H^2) exchange; Lys, L-lysine; PDB, Protein Data Bank; PLIMSTEX, protein−ligand interactions in solution monitored by MS, titration, and H/D exchange; Pyr, pyruvate; rmsd, root-mean-square deviation.

REFERENCES

(1) Wolfenden, R., and Snider, M. J. (2001) The depth of chemical time and the power of enzymes as catalysts. *Acc. Chem. Res. 34*, 938−945.

(2) Cleland, W. W., and Northrop, D. B. (1999) Energetics of Substrate Binding, Catalysis, and Product Release. *Methods Enzymol. 308*, 3−27.

(3) Henzler-Wildman, K. A., Thai, V., Lei, M., Ott, M., Wolf-Watz, M., Fenn, T., Pozharski, E., Wilson, A. M., Petsko, A. G., Karplus, M., Hübner, C. G., and Kern, D. (2007) Intrinsic motions along an enzymatic reaction trajectory. *Nature 450*, 838−844.

(4) Bhabha, G., Lee, J., Ekiert, D. C., Gam, J., Wilson, I. A., Dyson, H. J., Benkovic, S. J., and Wright, P. E. (2011) A Dynamic Knockout Reveals That Conformational Fluctuations Influence the Chemical Step of Enzyme Catalysis. *Science 332*, 234−238.

(5) Hammes-Schiffer, S. (2013) Catalytic Efficiency of Enzymes: A Theoretical Analysis. *Biochemistry 52*, 2012−2020.

(6) Liuni, P., Jeganathan, A., and Wilson, D. J. (2012) Conformer Selection and Intensified Dynamics During Catalytic Turnover in Chymotrypsin. *Angew. Chem., Int. Ed. 51*, 9666−9669.

(7) Liang, Z.-X., Lee, T., Resing, K. A., Ahn, N. G., and Klinman, J. P. (2004) Thermal-activated protein mobility and its correlation with catalysis in thermophilic alcohol dehydrogenase. *Proc. Natl. Acad. Sci. U. S. A. 101*, 9556−9561.

(8) Frauenfelder, H., Chen, G., Berendzen, J., Fenimore, P. W., Jansson, H., McMahon, B. H., Stroe, I. R., Swenson, J., and Young, R. D. (2009) A unified model of protein dynamics. *Proc. Natl. Acad. Sci. U. S. A. 106*, 5129−5134.

(9) Sauter, N. K., Echols, N., Adams, P. D., Zwart, P. H., Kern, J., Brewster, A. S., Koroidov, S., Alonso-Mori, R., Zouni, A., Messinger, J., Bergmann, U., Yano, J., and Yachandra, V. K. (2016) No observable conformational changes in PSII. *Nature 533*, E1−E2.

(10) Gebauer, F., and Hentze, M. W. (2004) Molecular mechanisms of translational control. *Nat. Rev. Mol. Cell Biol. 5*, 827−835.

(11) Deribe, Y. L., Pawson, T., and Dikic, I. (2010) Post-translational modifications in signal integration. *Nat. Struct. Mol. Biol. 17*, 666−672.

DOI: 10.1021/acs.biochem.6b00764
Biochemistry 2016, 55, 5413−5422

Downloaded from pubs.acs.org/biochaw/article-pdf/55/38/5413/7885135/bi6b00764.pdf by HARVARD UNIV user on 09 November 2026

(12) De Smet, F., Christopoulos, A., and Carmeliet, P. (2014) Allosteric targeting of receptor tyrosine kinases. *Nat. Biotechnol. 32*, 1113−1120.
(13) Nussinov, R., and Tsai, C. J. (2014) Unraveling structural mechanisms of allosteric drug action. *Trends Pharmacol. Sci. 35*, 256−264.
(14) Motlagh, H. N., Wrabl, J. O., Li, J., and Hilser, V. J. (2014) The ensemble nature of allostery. *Nature 508*, 331−339.
(15) Lee, S., Wales, T. E., Escudero, S., Cohen, D. T., Luccarelli, J., Gallagher, C., Cohen, N. A., Huhn, J., Bird, G. H., Engen, J. R., and Walensky, L. D. (2016) Allosteric Inhibition of Anti-Apoptotic MCL-1. *Nat. Struct. Mol. Biol. 23*, 600.
(16) Monod, J., Wyman, J., and Changeux, J. P. (1965) On the Nature of Allosteric Transitions - A Plausible Model. *J. Mol. Biol. 12*, 88−118.
(17) Schirmer, T., and Evans, P. R. (1990) Structural Basis of the Allosteric Behavior of Phosphofructokinase. *Nature 343*, 140−145.
(18) Dyachenko, A., Gruber, R., Shimon, L., Horovitz, A., and Sharon, M. (2013) Allosteric mechanisms can be distinguished using structural mass spectrometry. *Proc. Natl. Acad. Sci. U. S. A. 110*, 7235−7239.
(19) Ascenzi, P., Bocedi, A., Bolli, A., Fasano, M., Notari, S., and Polticelli, F. (2005) Allosteric modulation of monomeric proteins. *Biochem. Mol. Biol. Educ. 33*, 169−176.
(20) Verespy, S., III, Mehta, A. Y., Afosah, D., Al-Horani, R. A., and Desai, U. R. (2016) Allosteric Partial Inhibition of Monomeric Proteases. Sulfated Coumarins Induce Regulation, not just Inhibition, of Thrombin. *Sci. Rep. 6*, 24043.
(21) Eaton, W. A., Henry, E. R., Hofrichter, J., and Mozzarelli, A. (1999) Is cooperative oxygen binding by hemoglobin really understood. *Nat. Struct. Biol. 6*, 351−358.
(22) Bellelli, A., and Brunori, M. (2011) Hemoglobin allostery: Variations on the theme. *Biochim. Biophys. Acta, Bioenerg. 1807*, 1262−1272.
(23) Perutz, M. F., Wilkinson, A. J., Paoli, M., and Dodson, G. G. (1998) The stereochemical mechanism of the cooperative effects in hemoglobin revisited. *Annu. Rev. Biophys. Biomol. Struct. 27*, 1−34.
(24) Braun, A. P., and Schriemer, D. C. (2014) Tracking Allosteric Propagation with HX-MS. *Structure 22*, 512−514.
(25) Laskowski, R. A., Gerick, F., and Thornton, J. M. (2009) The structural basis of allosteric regulation in proteins. *FEBS Lett. 583*, 1692−1698.
(26) Hutton, C. A., Perugini, M. A., and Gerrard, J. A. (2007) Inhibition of lysine biosynthesis: an evolving antibiotic strategy. *Mol. BioSyst. 3*, 458−465.
(27) Skovpen, Y. V., and Palmer, D. R. J. (2013) Dihydrodipicolinate Synthase from Campylobacter jejuni: Kinetic Mechanism of Cooperative Allosteric Inhibition and Inhibitor-Induced Substrate Cooperativity. *Biochemistry 52*, 5454−5462.
(28) Cox, R. J. (1996) The DAP pathway to lysine as a target for antimicrobial agents. *Nat. Prod. Rep. 13*, 29−43.
(29) Skovpen, Y. V., Conly, C. J. T., Sanders, D. A. R., and Palmer, D. R. J. (2016) Biomimetic Design Results in a Potent Allosteric Inhibitor of Dihydrodipicolinate Synthase from Campylobacter jejuni. *J. Am. Chem. Soc. 138*, 2014−2020.
(30) Atkinson, S. C., Dogovski, C., Downton, M. T., Czabotar, P. E., Dobson, R. C., Gerrard, J. A., Wagner, J., and Perugini, M. A. (2013) Structural, kinetic and computational investigation of Vitis vinifera DHDPS reveals new insight into the mechanism of lysine-mediated allosteric inhibition. *Plant Mol. Biol. 81*, 431−446.
(31) Kumpaisal, R., Hashimoto, T., and Yamada, Y. (1987) Purification and characterization of dihydrodipicolinate synthase from wheat suspension cultures. *Plant Physiol. 85*, 145−151.
(32) Conly, C. J. T., Skovpen, Y. V., Li, S., Palmer, D. R. J., and Sanders, D. A. R. (2014) Tyrosine 110 Plays a Critical Role in Regulating the Allosteric Inhibition of Campylobacter jejuni Dihydrodipicolinate Synthase by Lysine. *Biochemistry 53*, 7396−7406.
(33) Young, K. T., Davis, L. M., and DiRita, V. J. (2007) *Campylobacter jejuni*: molecular biology and pathogenesis. *Nat. Rev. Microbiol. 5*, 665−679.
(34) Dobson, R. C. J., Valegard, K., and Gerrard, J. A. (2004) The crystal structure of three site-directed mutants of Escherichia coli dihydrodipicolinate synthase: Further evidence for a catalytic triad. *J. Mol. Biol. 338*, 329−339.
(35) Katz, B. A., Johnson, C., and Cass, R. T. (1995) Structure-based design of high affinity streptavidin binding cyclic peptide ligands containing thioether crosslinks. *J. Am. Chem. Soc. 117*, 8541−8547.
(36) Duncan, K. E., Dempsey, B. R., Killip, L. E., Adams, J., Bailey, M. L., Lajoie, G. A., Litchfield, D. W., Brandl, C. J., Shaw, G. S., and Shilton, B. H. (2011) Discovery and Characterization of a Non-phosphorylated Cyclic Peptide Inhibitor of the Peptidylprolyl Isomerase, Pin1. *J. Med. Chem. 54*, 3854−3865.
(37) Xiao, H., and Kaltashov, I. A. (2005) Transient Structural Disorder as a Facilitator of Protein-Ligand Binding: Native H/D Exchange—Mass Spectrometry Study of Cellular Retinoic Acid Binding Protein I. *J. Am. Soc. Mass Spectrom. 16*, 869−879.
(38) Shadrina, M. S., Peslherbe, G. H., and English, A. M. (2015) Quaternary-Linked Changes in Structure and Dynamics That Modulate O-2 Migration within Hemoglobin's Gas Diffusion Tunnels. *Biochemistry 54*, 5268−5278.
(39) Englander, S. W., Mayne, L., and Krishna, M. M. G. (2007) Protein folding and misfolding: mechanism and principles. *Q. Rev. Biophys. 40*, 287−326.
(40) Iacob, R. E., and Engen, J. R. (2012) Hydrogen Exchange Mass Spectrometry: Are We Out of the Quicksand? *J. Am. Soc. Mass Spectrom. 23*, 1003−1010.
(41) Rand, K. D., Zehl, M., and Jorgensen, T. J. D. (2014) Measuring the Hydrogen/Deuterium Exchange of Proteins at High Spatial Resolution by Mass Spectrometry: Overcoming Gas-Phase Hydrogen/Deuterium Scrambling. *Acc. Chem. Res. 47*, 3018−3027.
(42) Konermann, L., Pan, J., and Liu, Y. (2011) Hydrogen Exchange Mass Spectrometry for Studying Protein Structure and Dynamics. *Chem. Soc. Rev. 40*, 1224−1234.
(43) Underbakke, E. S., Iavarone, A. T., Chalmers, M. J., Pascal, B. D., Novick, S., Griffin, P. R., and Marletta, M. A. (2014) Nitric Oxide-Induced Conformational Changes in Soluble Guanylate Cyclase. *Structure 22*, 602−611.
(44) Sowole, M. A., and Konermann, L. (2014) Effects of Protein-Ligand Interactions on Hydrogen/Deuterium Exchange Kinetics: Canonical and Non-Canonical Scenarios. *Anal. Chem. 86*, 6715−6722.
(45) Deredge, D., Li, J. W., Johnson, K. A., and Wintrode, P. L. (2016) Hydrogen/Deuterium Exchange Kinetics Demonstrate Long Range Allosteric Effects of Thumb Site 2 Inhibitors of Hepatitis C Viral RNA-dependent RNA Polymerase. *J. Biol. Chem. 291*, 10078−10088.
(46) Donovan, K. A., Zhu, S. L., Liuni, P., Peng, F., Kessans, S. A., Wilson, D. J., and Dobson, R. C. J. (2016) Conformational Dynamics and Allostery in Pyruvate Kinase. *J. Biol. Chem. 291*, 9244−9256.
(47) Arora, J., Hickey, J. M., Majumdar, R., Esfandiary, R., Bishop, S. M., Samra, H. S., Middaugh, C. R., Weis, D. D., and Volkin, D. B. (2015) Hydrogen exchange mass spectrometry reveals protein interfaces and distant dynamic coupling effects during the reversible self-association of an IgG1 monoclonal antibody. *mAbs 7*, 525−539.
(48) Beveridge, R., Migas, L. G., Payne, K. A. P., Scrutton, N. S., Leys, D., and Barran, P. E. (2016) Mass spectrometry locates local and allosteric conformational changes that occur on cofactor binding. *Nat. Commun. 7*, 12163.
(49) Wales, T. E., Fadgen, K. E., Gerhardt, G. C., and Engen, J. R. (2008) High-Speed and High-Resolution UPLC Separation at Zero Degree Celsius. *Anal. Chem. 80*, 6815−6820.
(50) Weis, D. D., Engen, J. R., and Kass, I. J. (2006) Semi-automated data processing of hydrogen exchange mass spectra using HX-Express. *J. Am. Soc. Mass Spectrom. 17*, 1700−1703.
(51) Zhu, M. M., Rempel, D. L., Du, Z. H., and Gross, M. L. (2003) Quantification of protein-ligand interactions by mass spectrometry,

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DOI: 10.1021/acs.biochem.6b00764
Biochemistry 2016, 55, 5413−5422

titration, and H/D exchange: PLIMSTEX. *J. Am. Chem. Soc. 125*, 5252−5253.
(52) Frauenfelder, H. (2014) Ask not what physics can do for biology-ask what biology can do for physics. *Phys. Biol. 11*, 053004.
(53) Osapay, K., Theriault, Y., Wright, P. E., and Case, D. A. (1994) Solution structure of carbonmonoxy myoglobin determined from nuclear magnetic resonance distance and chemical shift constraints. *J. Mol. Biol. 244*, 183−197.
(54) Dobson, R. C. J., Devenish, S. R. A., Turner, L. A., Clifford, V. R., Pearce, F. G., Jameson, G. B., and Gerrard, J. A. (2005) Role of arginine 138 in the catalysis and regulation of Escherichia coli dihydrodipicolinate synthase. *Biochemistry 44*, 13007−13013.
(55) Tajkhorshid, E., Nollert, P., Jensen, M. Ø., Miercke, L. J. W., O'Connell, J., Stroud, R. M., and Schulten, K. (2002) Control of the Selectivity of the Aquaporin Water Channel Family by Global Orientational Tuning. *Science 296*, 525−530.

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